

TITLE & AUTHOR INFORMATION

Air Particulate Matter Exposure is Associated With Non-Shockable Cardiac Rhythm in Out-of-Hospital Cardiac Arrest

Scott Cohen, Emmett Martin, MPH, Muhammed AB Chowdhury, MPH, Joseph A Tyndall, MD, MPH, Torben K Becker, MD, PhD

Department of Epidemiology & Department of Emergency Medicine,
College of Medicine, University of Florida

STUDY OVERVIEW

BACKGROUND

- Death and disability from out-of-hospital cardiac arrest (OHCA) remains a prominent public health threat in the United States
- Air particulate matter less than 2.5 microns (PM2.5) is an independent risk factor for cardiovascular mortality and morbidity
- Previous studies have reported the association between short-term PM2.5 exposure and the risk of out-of-hospital cardiac arrest, however its relationship to the cardiac rhythm at OHCA onset is unknown.

OBJECTIVE

- We investigated the relationship between location-controlled, short-term and acute PM2.5 exposure and initial cardiac rhythm in OHCA

STUDY DESIGN

- In this retrospective cohort study, emergency medical service records of adult, non-traumatic OHCA within Alachua County, FL from 2012-2014 were reviewed and matched with UF Health electronic records
- Patients who did not survive to the emergency department or unable to be geocoded were excluded to avoid variable misclassification (**Figure 1**)
- Locations of cardiac arrest incidence were geocoded and spatially joined to corresponding census tract (ArcGIS) (**Figure 2**)
- Daily, census tract-level PM2.5 measurements 0, 1, 2, and 3 days before event were assigned to each subject

Primary Exposure

Daily, census tract-level fine particulate matter (PM2.5)

CDC Public Health Tracking Network

Primary Outcome

Presenting cardiac rhythm determined by EMS on arrival

Non-Shockable vs. Shockable Rhythms

STATISTICAL ANALYSIS

- Descriptive statistics were applied to assess cohort by outcome (**Table 1**)
- Wilcoxon rank-sum test was used to compare exposure by outcome (**Figure 3**)
- Multilevel mixed-effects logistic regression model was developed to assess the relationship between PM2.5 exposure and the probability of non-shockable initial cardiac rhythm (**Figure 4, Figure 5**)
- Intraclass correlation coefficient (ICC) was evaluated to assess the geographic and temporal variation of exposure/outcome (**Table 2**)

CONCLUSION

- Short-term, location-controlled PM2.5 exposure is associated with a less desirable initial cardiac rhythm in OHCA
- This research suggests that short-term fine particulate matter exposure may be an indicator or modifier of OHCA etiology.

Short-term air pollution exposure increases odds of non-shockable cardiac rhythm during out-of-hospital cardiac arrest



Society of Academic
Emergency Medicine
May 14th, 2020

TABLES & FIGURES

Figure 1: Study inclusion flow diagram

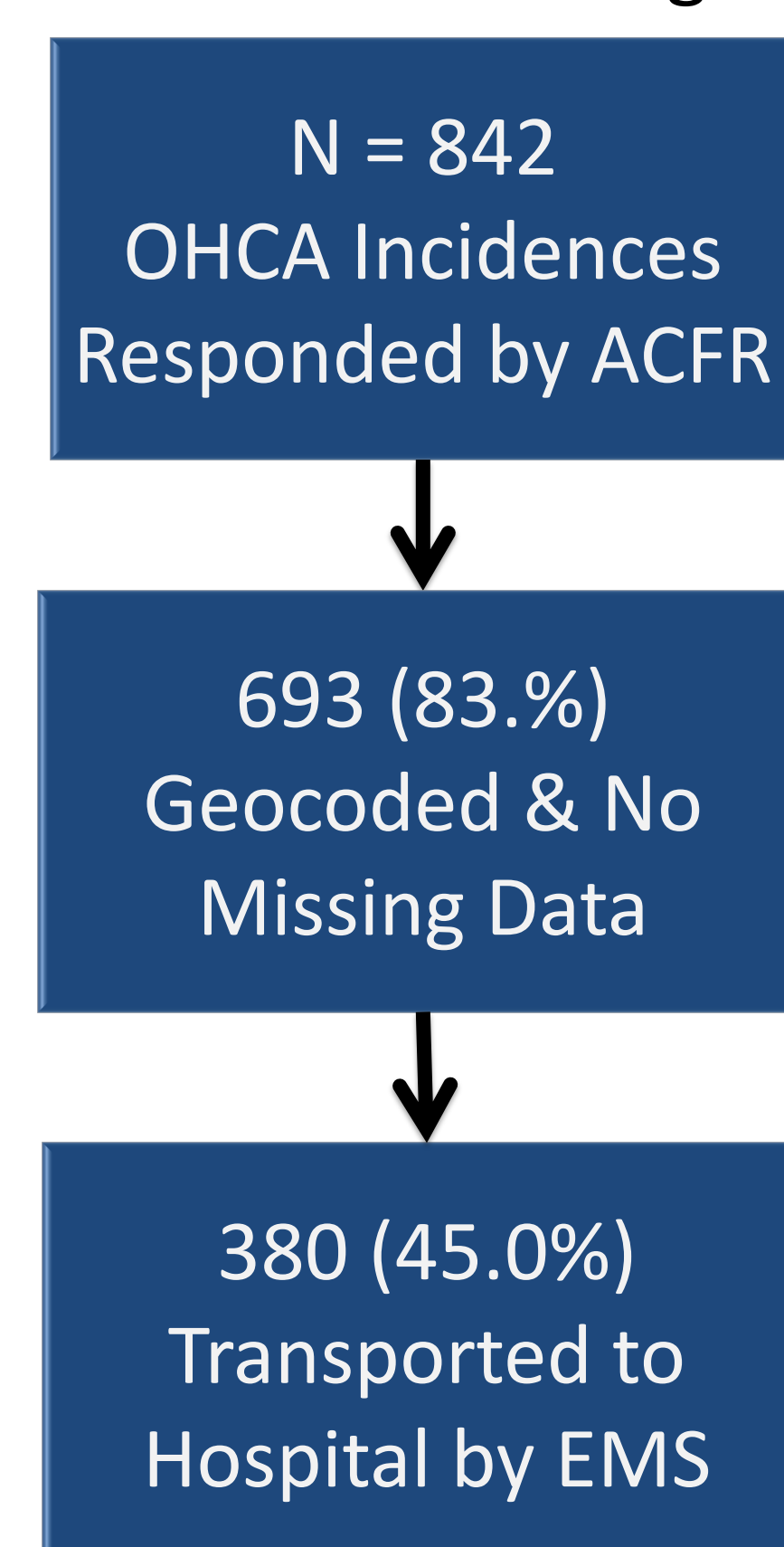


Figure 2: Heat map of population-adjusted spatial density of OHCA incidence (OHCA/km²/year) [Range: 0 – 60]

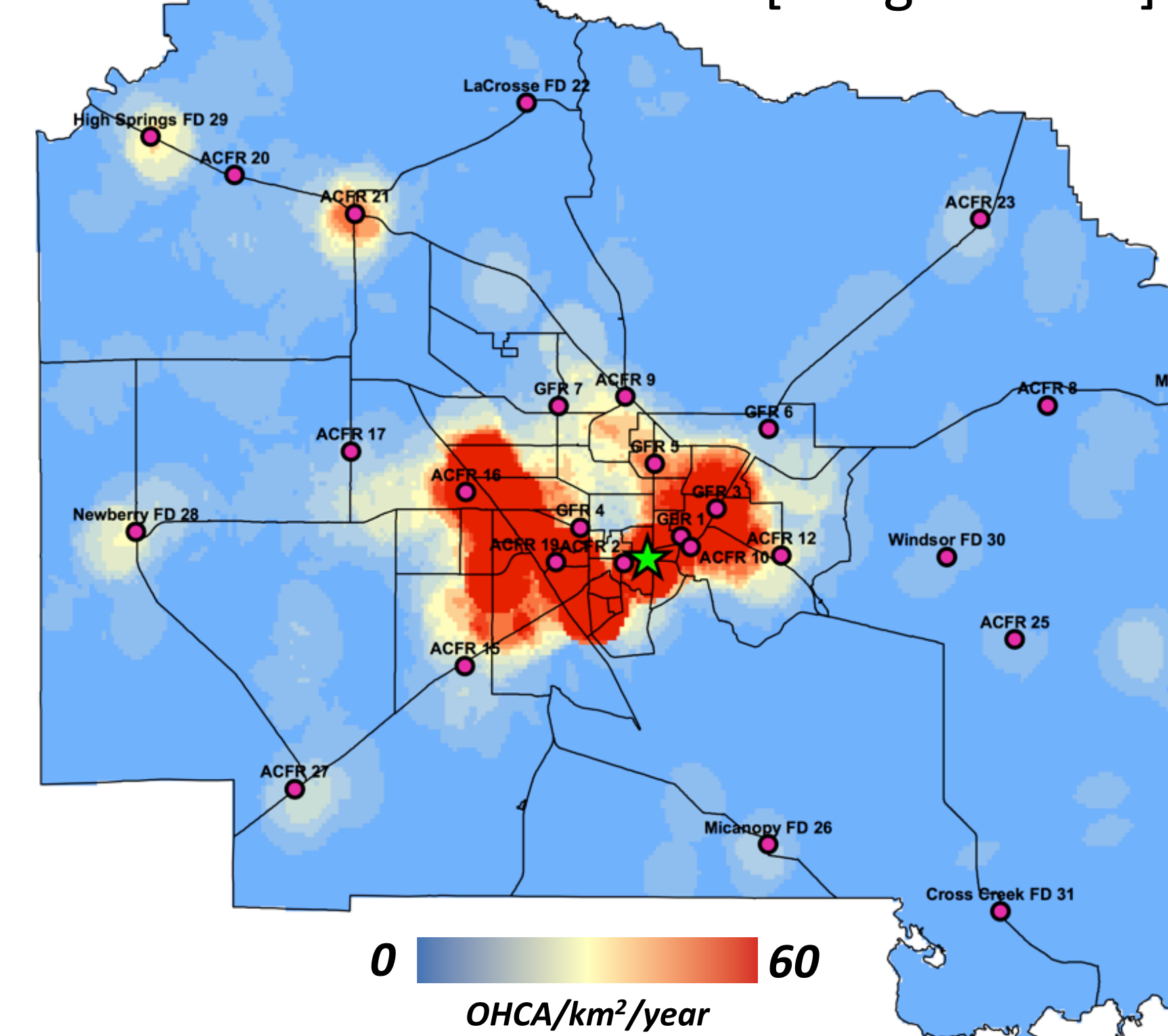


Figure 3: Particulate matter exposure by initial cardiac rhythm on day of event

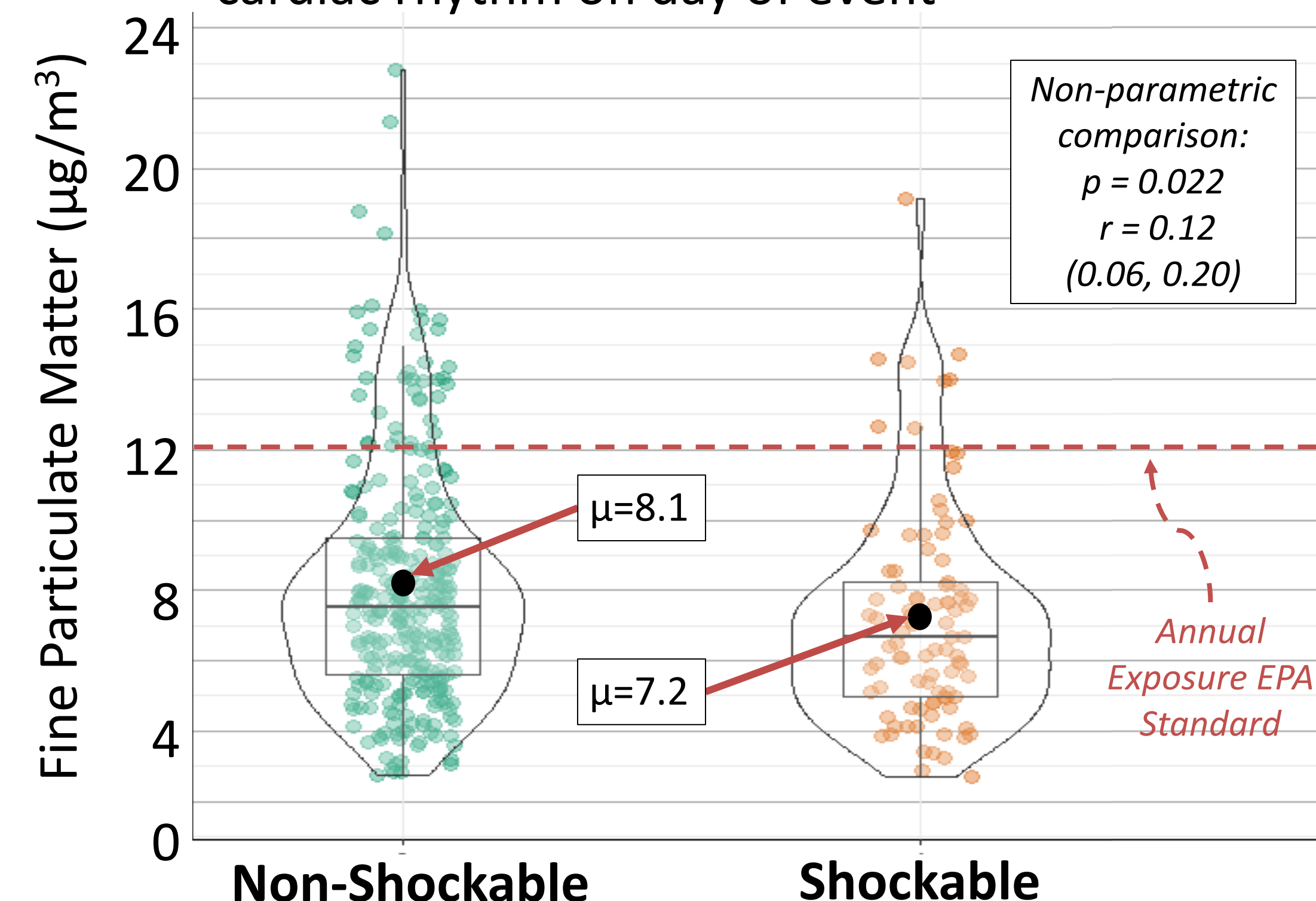


Table 2: Assessment of geographic and temporal variation in exposure/outcome using multilevel models

Cluster Variation ICC (P-value)	Particulate Matter Exposure	Non-Shockable Probability
Geography	2.2% (0.229)	11.3% (0.036)
Day of Week of Event	2.0% (0.171)	0.0% (0.999)
Week of Event	17.9% (0.002)	0.0% (0.999)
Month of Event	12.6% (0.022)	0.0% (0.999)
Season of Event	8.3% (0.103)	0.0% (0.999)

Table 1: Study demographics, predictors, and outcomes of eligible OHCA cohort by initial cardiac rhythm

Patient Predictors	Non-Shockable (n=287, 75.5%)	Shockable (n=93, 24.5%)	P-value
Age, mean (SD)	64.9 (17.4)	66.0 (16.2)	0.574
Sex (% Female)	136 (47.4)	27 (29.0)	0.003
Race:			
Black/African-Amer.	102 (35.5)	35 (26.2)	0.229
White	184 (64.1)	96 (72.4)	
Other/Unknown	1 (0.4)	18 (1.4)	
Bystander AED:	8 (4.3)	6 (11.5)	0.088
Bystander CPR:	91 (49.2)	33 (63.5)	0.084
Time of Day (AM vs. PM)	113 (39.4)	43 (41.9)	0.715
Survival to Discharge*	14 (10.5)	18 (40.9)	<0.001
Survival to Home Discharge	9 (6.7)	10 (22.7)	0.008

Figure 4: Odds ratios (OR) for having non-shockable rhythm at every 50% increase in daily PM2.5 exposure

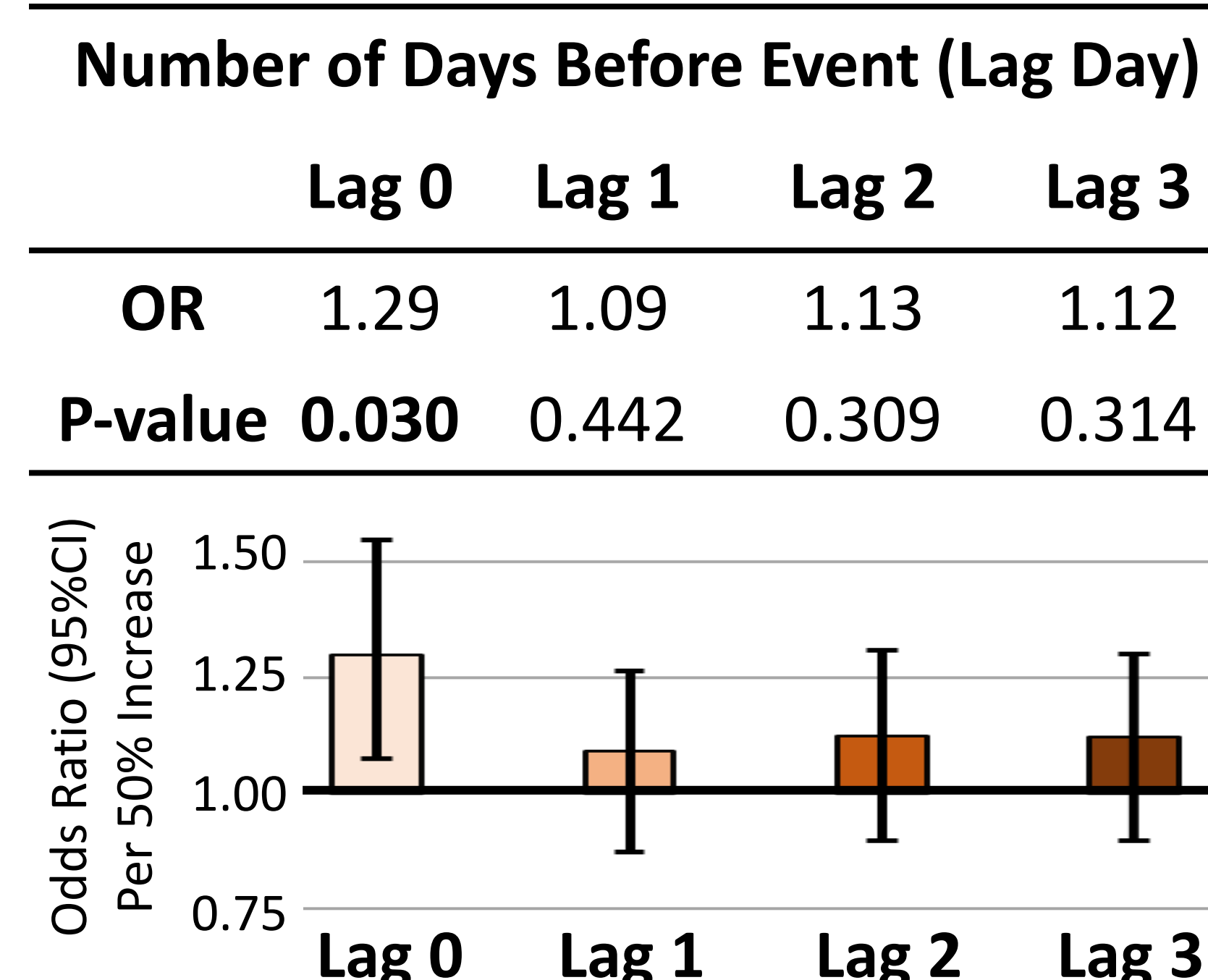
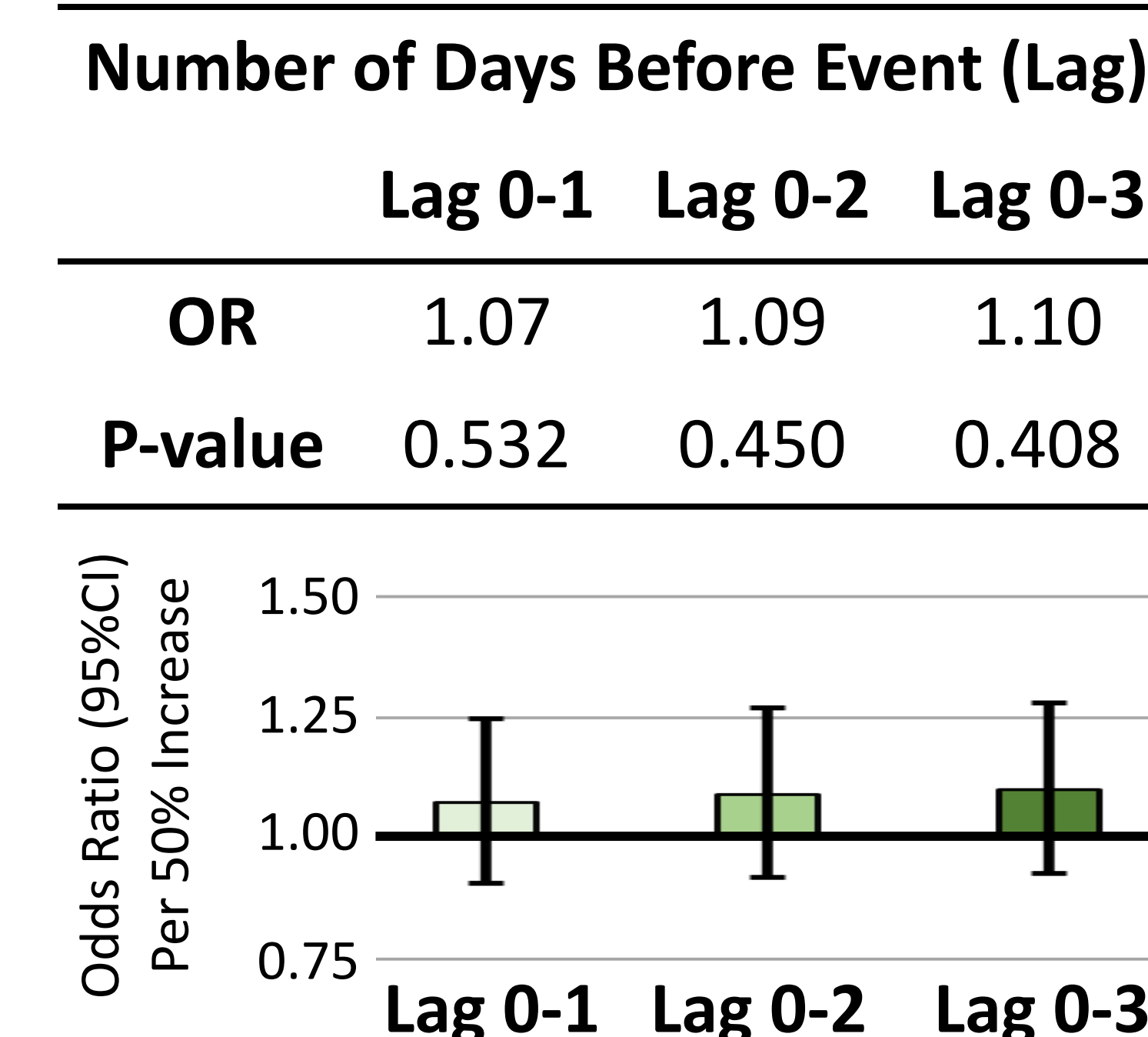
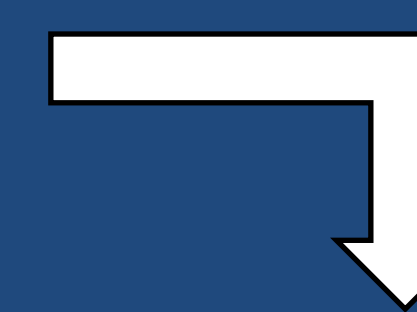


Figure 5: Odds ratio (OR) for having non-shockable rhythm at every 50% increase in average PM2.5 exposure



Learn How to
Assess for
Geographic
Disparities in
Cardiac Arrest
Survival in Your
County!
Scan for More!



UF Department of Epidemiology
UNIVERSITY of FLORIDA

UF Department of
Emergency Medicine
UNIVERSITY of FLORIDA